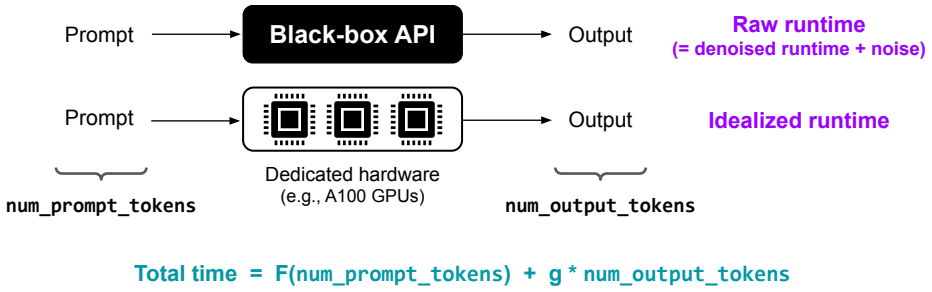

Figure 22: **Inference Efficiency Metrics.** A demonstration of how we measure inference efficiency. We compute two metrics: denoised inference runtime and idealized inference runtime. F returns the runtime of encoding a prompt of given size, and g is the runtime of generating each additional output token for the given model.

For some models, like the AI21 models, we do not have enough information to make a reliable estimate. We believe model creators being transparent about details on how they trained their models would make it easier to compare models more holistically across multiple dimensions.

4.9.2 Inference efficiency

For inference, we would ideally want to do something similar by reporting total emitted CO_2 or kWh for each inference request; this is not immediately tractable, however, since the hardware used to serve the requests is not public information.

One alternative is to report **per-request runtime**, which is what users using deployed systems experience in their applications. Per-request runtime, however, cannot be used to compare models and model providers due to disparities in how these models are served. For example, two model providers' deployments can differ on:

- Hardware: both type of accelerator and number of accelerators.
- Software implementations and optimizations.
- Amount of performance variation due to contention, which can lead to requests spending time in queues waiting for resources to become available as opposed to on computation.

These are unfortunately not *fundamental* to the model itself, and consequently do not allow us to compare models on a level footing, which is the objective of this work. To be able to compare models more fairly, we designed two metrics:

- **Denoised inference runtime.** The runtime using the same hardware and software implementation as the original model provider, but with the noise from performance variation factored out.
- **Idealized inference runtime.** The runtime using uniform *optimized* hardware and software implementation, allowing for the inference efficiency of models to be directly compared against each other.

We present both of these metrics since we believe both are important: denoised runtimes give us an estimate for how long queries will take for end users using deployed interfaces such as OpenAI's API in the *best case*, while idealized runtimes allow models to be more fairly compared and can be used to understand efficiency-capability tradeoffs. We present denoised runtimes for every model, and idealized runtimes for all models with publicly available model architecture information. In practice, we measure idealized runtimes on NVIDIA A100 GPUs using Megatron (Shoeybi et al., 2019), since we believe these are optimized hardware and software stacks (at the time of writing) with features like model parallelism that are needed to serve

large LMs. Figure 22 shows a visual comparison between these different inference runtime metrics, and also briefly explains how the denoised and idealized runtime metrics can be estimated.

We can also derive idealized energy and idealized emitted CO_2 metrics from the idealized runtime, since we control the hardware on which the idealized runtimes are estimated. Note that this cannot be done for the denoised runtimes since we do not know the hardware used by the model provider.

5 Targeted evaluations

In §3: CORE-SCENARIOS, we prioritized user-facing scenarios, where progress could confer direct social utility. Given the growing development and deployment of language models, holistic evaluation of model performance for such scenarios aims to track the impact models will have. However, as Bommasani et al. (2021, §4.4) describe, evaluation serves multiple functions and the relevant function can depend on the stakeholder (e.g. researchers have different evaluation needs from policymakers). While the preceding evaluation provides clarity on the practical utility of existing models, it is less effective at providing fine-grained scientific insight on primitives of interest. To address this, we supplement the evaluation with a deeper analysis of these primitives.

Akin to how we explored the structure latent in the space of scenarios systematically, we identify further structure in designating a set of *components* that help determine the benefits and harms of a language model. On the side of capabilities, we consider the canonical primitives of language, knowledge, and reasoning. On the side of harms, the space of harms of language models is more nascent, so we follow the recent taxonomies of Bommasani et al. (2021, §5), Weidinger et al. (2022), Bender et al. (2021) and Rauh et al. (2022). Specifically, we focus on disinformation and copyright as first-order concerns that are especially concerning when language models are accessible to malicious actors. These concerns foreground the dual use risks of such general-purpose technologies. Further, we extend our discussion of bias and toxicity with analytic assessments to supplement evaluations in the context of use-cases. Overall, our evaluation aims to build understanding on the practical utility of language models for society as well as the fundamental scientific components that shape model behavior.

5.1 Language

To measure a model's understanding of the English language, we evaluate it on two types of scenarios: language modeling and minimal pairs.[48] Both approaches have extensive traditions in linguistics: the former has been widely studied in incremental processing in psycholinguistics (Hale, 2001; Levy, 2008), where the performance at predicting the next word clarifies how well models or humans have learned the distribution of language use in a domain. The latter approach of minimal pair comparison is a dominant methodology employed throughout linguistics (Chomsky, 1957) that helps to tease out fine-grained understanding of specific linguistic phenomena. Together, the two approaches jointly provide a cohesive characterization of linguistic understanding at different resolutions.

5.1.1 Language modeling

Problem setting. In language modeling, the input is an English text sequence that the model assigns a conditional log probability for each token in the sequence that can be summed to assign a (log) probability to the full sequence. More precisely, since we compare models that use different tokenizers, we use bits per byte as our main metric because of its invariance to tokenization schemes following Gao et al. (2021a).

Datasets and selection process. Among all existing language modeling benchmarks, we select the following: **WikiText-103** (Merity et al., 2016), **The Pile** (Gao et al., 2021a), **TwitterAAE** (Blodgett et al., 2016), and **ICE** (The International Corpus of English; Greenbaum, 1991). We include **WikiText-103** as a long-standing and well-studied benchmark for language modeling that covers English Wikipedia data and **The Pile** for its broader domain coverage. **WikiText-103** is a collection of verified articles from

[48] We recognize the term *understanding* is contested and subject to several valid interpretations (see Bommasani et al., 2021, §2.6). We clarify our operationalization below.

Wikipedia used for language model benchmarking. **The Pile** is a compilation of 22 different sub-corpora, of which we prioritize 5 corpora to ensure coverage of different domains: arXiv, BookCorpus2, Enron Emails, PubMed Central, and Wikipedia (en).

To these, we add **TwitterAAE** and **ICE** to evaluate language modeling across a range of English varieties. These corpora, in addition to extending our coverage of English text for evaluation, serve to measure performance disparities in a model's understanding of different English varieties. Blodgett et al. (2020), taking African American English (AAE) as an example, argues that deficits in language technology performance for AAE are inherently harmful because they risk reinforcing a stigmatization of AAE that has historically and continues to deny social opportunity to the speakers of that language. Given the well-documented center-periphery domination between English as a Native Language (ENL) varieties and English as a Second/Foreign Language (ESL/EFL) varieties (Kachru et al., 2009), we expect that disparities in language modeling capacity between those two groups of regional Englishes could be similarly harmful.

TwitterAAE contains over 50 million social media messages from Twitter ("tweets") labeled with demographic proportions predicted using geolocations of the users. Similar to Blodgett et al. (2016), we sample 50,000 examples from the 830,000 tweets with the highest African American proportions and the 7.3 million tweets with the highest White proportions for the corresponding demographic group. **ICE** is a set of corpora designed for comparative analysis of 12 regional/national varieties of English across the world. Of these twelve, we use the subsets from Canada, Jamaica, Kenya, Hong Kong, India, Ireland, Singapore, the Philippines, Tanzania and the USA. These, in line with the two dominant classifications of world Englishes—the Three Circles model and the Dynamic model (Kirkpatrick, 2020)—constitute a representative grouping of English varieties.

5.1.2 Minimal pairs

Problem setting. A *minimal pair* is a pair of sequences that differ in a single token, where one sequence is acceptable and the other is not. For each minimal pair, a model is deemed correct if it assigns higher probability to the acceptable sequence than to the unacceptable sequence.

Datasets and selection process. Minimal pair evaluation of language models was pioneered by Linzen et al. (2016), leading to several works that amassed minimal pairs to test linguistic understanding (Linzen et al., 2016; Warstadt et al., 2020; Gauthier et al., 2020, *inter alia*). Of these, we elect to use BLiMP (Benchmark for Linguistic Minimal Pairs; Warstadt et al., 2020), which contains minimal pairs testing syntactic, morphological, and semantic knowledge. Drawing from syntax textbooks in linguistics to converge on a set of core phenomena, it covers 12 linguistic phenomena and 67 paradigms where 1000 synthetic minimal pairs were programmatically generated for each paradigm.

5.2 Knowledge

To measure a model's knowledge, we evaluate the model through question answering (§5.2.1: KNOWLEDGE-QA) and text completion (§5.2.2: KNOWLEDGE-FACTCOMPLETION).[49] Question answering enables us to re-use questions developed for assessing human knowledge (e.g. academic exams) that involve light linguistic understanding and reasoning, whereas text completion allows us to further isolate specific factual knowledge.

5.2.1 Knowledge-intensive QA

To evaluate LMs' knowledge in a practical QA setting, we use existing QA benchmarks that require significant knowledge to solve.

Datasets. Among the QA datasets discussed in §3.3: QUESTIONANSWERING, we focus on a subset that tests diverse forms of knowledge: **HellaSwag**, **OpenBookQA**, **TruthfulQA**, and **MMLU**. Of these, **HellaSwag** and **OpenBookQA** test general commonsense knowledge, with **TruthfulQA** further adding

[49]To be more precise, since we are evaluating models as text-to-text interfaces, this means knowledge that is currently used by the interface. In other words, the knowledge need not be stored in the model and could, for example, be stored in an external knowledge base that is accessed by a retrieval-augmented model (Lewis et al., 2020c; Yasunaga et al., 2021; 2022a).